



Regular Article

ReliefNet: A Knowledge-Driven, Explainable AI Multimodal Framework for Disaster Severity Classification and Humanitarian Decision-Making

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ARTICLE INFO

Keywords:

Deep Learning
Multimodal Learning
Transformer
Explainable AI (XAI)
CLIP
Disaster

ABSTRACT

Rapid and reliable disaster severity assessment from social media is difficult due to noisy content, modality imbalance, and limited labeled data. This paper introduces ReliefNet, a real-time multimodal framework for disaster severity classification that jointly analyzes images and text. The system integrates DualEmbedNet, a dual-encoder transformer for textual classification, and DisasterNet, a CNN with channel-wise attention for visual analysis, combined through an accuracy-weighted decision-level fusion mechanism that dynamically adapts to modality reliability at inference time. To mitigate labeling scarcity, we construct a unified dataset of 14,996 image-text pairs by merging CrisisMMD and TSEqD. Unlabeled samples are annotated using unsupervised multimodal K-Means clustering, followed by Grad-CAM guided ROI refinement and manual auditing to reduce label noise. Models are evaluated using an 80/10/10 train-validation-test split with metrics including accuracy, precision, recall, F1-score, calibration error, and robustness under missing-modality conditions. The image-only model achieves an F1-score of 97.66%, while ReliefNet attains 97.00% F1 with superior robustness and stability under noisy or incomplete inputs. Accuracy-weighted fusion improves F1-score by 0.7% over equal-weight fusion and outperforms standard CNN baselines such as ResNet and EfficientNet. All models operate in real time on a single NVIDIA GPU, demonstrating ReliefNet's practical deployability.

1. Introduction

Natural catastrophes such as earthquakes, floods, and hurricanes are increasing in frequency and intensity, causing widespread socio-economic disruption. These events impose severe losses on infrastructure, public health systems, livelihoods, and national economies [1,2]. Recent disasters illustrate the scale of this challenge. In 2024, Super Typhoon Yagi, the strongest storm to strike Vietnam in seven decades, caused extensive flooding and landslides across Southeast Asia, resulting in nearly 600 fatalities and affecting more than 5.6 million people [3]. In the same year, Hurricane Helene triggered large-scale flooding and infrastructure failure in the United States, causing at least 219 deaths and becoming the deadliest Atlantic hurricane to strike the mainland since Hurricane Katrina [4]. Such events strain emergency response systems and expose the limitations of existing disaster assessment

mechanisms, particularly in capturing cascading social and economic impacts beyond immediate physical damage [5–7].

At the same time, social media has emerged as a critical information source during crisis events. Affected populations increasingly share real-time observations, images, and requests for assistance through online platforms. These posts are inherently multimodal, combining text and visual content, and play an important role in situational awareness, rescue coordination, and humanitarian response [1,8]. The immediacy of social media data offers clear advantages for crisis management, but the sheer volume of content, much of it irrelevant or misleading, makes manual analysis infeasible [9]. Automated methods are therefore required to identify informative content and assess disaster severity at scale [10,11]. Text and images alone are insufficient: crisis-related text is often informal, noisy, and ambiguous, while images without contextual grounding may be misinterpreted. Effective disaster assessment

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<https://doi.org/10.1016/j.pdisas.2026.100528>

Received 16 October 2025; Received in revised form 25 December 2025; Accepted 25 January 2026

Available online 3 February 2026

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thus depends on integrating complementary information from both modalities [12,13].

Despite progress in multimodal disaster classification, several limitations remain. Many existing approaches [13–20] effectively combine visual and textual features using Convolutional Neural Networks (CNNs), recurrent models, or transformer-based architectures. These methods have demonstrated the strong potential of multimodal learning for disaster-related content analysis, while leaving opportunities for further exploration of richer cross-modal interactions to better leverage complementary cues across modalities. Social context signals related to credibility or user engagement are rarely incorporated, despite their relevance for filtering misinformation. Noise and semantic inconsistency arising from misspellings, hashtags, reused images, and informal expressions remain persistent challenges. In operational settings, unimodal pipelines are still widely used, even though they cannot capture the full situational context of disaster events. Recent advances such as Contrastive Language-Image Pre-Training (CLIP)-style joint vision–language embeddings, systematic optimization strategies for high-dimensional crisis data, and interpretable evaluation pipelines have not yet been widely adopted for disaster severity classification. Limited transparency and insufficient external validation further reduce trust in real-time, high-stakes deployment scenarios.

Several emerging directions in multimodal learning offer opportunities to address these shortcomings but remain underexplored in disaster applications. Open-world recognition models capable of handling unseen disaster categories have not been systematically evaluated on crisis datasets. Graph-based approaches that model spatial, temporal, or social relationships have seen limited use in severity assessment. While explainable AI (XAI) methods have been adopted in domains such as healthcare and finance, their application to disaster severity classification remains limited, leaving emergency responders with little insight into the basis of model predictions. This study addresses the following research questions:

1. How can textual and visual data from social media be effectively integrated to improve disaster severity classification compared with unimodal approaches?
2. How can disaster severity be quantified from multimodal social media content in a manner suitable for time-sensitive decision support?
3. Which architectural choices best balance multimodal modeling capacity with computational efficiency for real-time deployment?

To address these questions, we propose ReliefNet, a real-time, explainable multimodal framework for disaster severity classification. ReliefNet combines CLIP, Bidirectional Encoder Representations from Transformers (BERT), and Residual Network (ResNet) representations to capture both aligned and complementary information across text and images. A decision-level fusion strategy preserves modality specific strengths while maintaining reliable performance under missing-modality conditions. Gradient-weighted Class Activation Mapping (Grad-CAM) guided data curation supports data quality assurance and interpretability. The framework is evaluated with respect to robustness under non-IID conditions, external generalization, and computational efficiency for real-time use. The key contributions of this work are summarized as follows:

1. We combined the CrisisMMD and Turkey–Syria Earthquake Dataset (TSEqD) into a single multimodal dataset containing 14,996 labeled samples after preprocessing. This makes it one of the largest datasets available for disaster severity classification (Table 4, 7).
2. Organized the previously unstructured TSEqD dataset into three disaster severity levels using a multimodal clustering pipeline based on ResNet50, BERT, CLIP, and K-Means clustering, followed by manual verification. This is the first work to provide structured severity labels for TSEqD (Table 4, 5, 7).

3. We applied XAI methods Grad-CAM, Grad-CAM++, LIME, and SHAP to understand why the models make certain predictions. These techniques were used to analyze both correct and incorrect predictions, improving the interpretability and trustworthiness of the system (Fig. 8, 9, 10, 11).
4. Introduced DualEmbedNet, a text model that uses a dual-embedding strategy to better capture disaster severity related language patterns in social media posts and tweets (Fig. 4).
5. Introduced **DisasterNet**, an image model that leverages hierarchical visual features to estimate disaster severity from images (Fig. 5).
6. We proposed ReliefNet, a multimodal framework that combines text, image, and joint embeddings. ReliefNet consistently outperforms individual text-only and image-only models, leading to more reliable disaster severity classification (Fig. 6).
7. Implemented a lightweight, real-time detection framework designed for fast inference, making it suitable for practical deployment in time-critical emergency response scenarios (Fig. 12, 13).

The remainder of this paper is organized as follows. Section 1.1 reviews related work. Section 2 describes the proposed methodology. Section 3 presents experimental results and analysis. Section 4 concludes with limitations and directions for future research.

1.1. Recent studies

Social media has become a primary source for Situational Awareness (SA) during disasters, enabling rapid information dissemination and damage reporting [15,16]. Accordingly, disaster-related studies increasingly focus on automatically categorizing crisis content to support emergency response. Existing approaches can be broadly grouped into unimodal methods, which analyze either text or images independently, and multimodal methods, which integrate multiple data types to improve severity assessment.

1.2. Unimodal approach

1.2.1. Text-Based approach

Text-based disaster analysis focuses on extracting severity-related signals from social media posts using natural language processing and Machine Learning (ML) techniques. Recent work demonstrates that transformer-based architectures dominate this area, particularly for

Table 1
Summary of recent text-based approaches.

Model / Approach	Year	Task Focus	Key Performance	Ref.
DistilBERT + NER	2023	Text classification & location extraction	99% text accuracy, 89% location accuracy	[17]
PoS Tagging	2024	Linguistic analysis for severity prioritization	Near real-time severe area identification	[18]
Sentiment Analysis	2023	Sentiment analysis in disaster tweets	5–7% accuracy gain over baselines	[19]
Topic Modeling	2023	Topic discovery & damage severity analysis	Effective identification of critical areas	[20]
BERT-based Models	2024	Disaster tweet classification	95% accuracy	[21]
CrisisTransformers	2024	Crisis-specific text representation	+17.43% embedding improvement	[22]
Hybrid Multimodal Model	2024	Joint image–text disaster classification	Superior event classification performance	[23]
Random Forest	2024	Flood severity classification	78% accuracy	[24]
XGBoost	2024	Flood severity classification	68% accuracy	[24]

disaster tweet classification and damage-related information extraction (Table 1). BERT and its variants consistently report strong performance on disaster datasets, achieving up to 95–99% accuracy when combined with auxiliary components such as Named Entity Recognition (NER) for identifying key entities such as locations and casualties [21,22]. Crisis-specific pretraining further improves performance; CrisisTransformers report substantial gains in sentence representation quality compared to general-purpose language models [23], underscoring the importance of domain adaptation in crisis contexts.

Complementary linguistic techniques are often used to enhance severity understanding. NER has been applied to flood-related posts to identify damage indicators with up to 89% accuracy [24], while part-of-speech tagging supports rapid prioritization of highly affected areas by capturing grammatical structure in short, informal text [25]. Sentiment analysis provides additional contextual signals, improving classification accuracy by 5–7% in disaster-related tweets [26], and topic modeling has been used to track evolving disaster themes over time [27]. Traditional ensemble models such as Random Forest (RF) and XGBoost remain relevant in low-resource settings, though they generally underperform transformer-based approaches (Table 1).

Despite these advances, text-only methods remain limited for severity assessment, as they fail to capture visual evidence of damage and are sensitive to informal language, multilingual usage, and rapidly changing discourse patterns. Models trained on narrow, event-specific datasets often exhibit reduced generalizability when deployed across disaster types or regions.

1.2.2. Image-based approach

Image-based disaster analysis leverages visual content from social media, drones, and satellite imagery to estimate damage severity and affected areas. Recent studies primarily rely on convolutional and transformer-based vision models tailored to disaster-specific tasks such as flooding detection [28], building damage assessment, and semantic segmentation (Table 2).

Within this domain, ResNet- and DenseNet-based architectures remain widely used due to their stability and strong performance on disaster imagery. DenseNet201 achieves over 99% accuracy on climate-related disaster datasets, while ResNet-50 and EfficientNet variants report average accuracies around 96% across multiple disaster classification benchmarks [25,26]. Ensemble strategies further improve

robustness; stacked CNN models combined with gradient-boosting classifiers achieve both high accuracy and class-level F1 scores [27]. Transformer-based vision models extend these capabilities to fine-grained damage assessment. SegFormer demonstrates strong segmentation performance on Unmanned Aerial Vehicle (UAV)-based disaster datasets such as RescueNet and FloodNet [29], while SLgViT integrates local and global visual cues for pixel-level building damage detection [34]. For real-time assessment, YOLOv8 enables rapid damage localization in drone imagery with high recall and precision [30].

Several studies address data scarcity and annotation challenges. Generative Adversarial Network (GAN)-based data augmentation improves validation accuracy by generating diverse disaster scenes [31], and pseudo-labeling frameworks such as DAVI reduce reliance on manually annotated ground truth while maintaining performance across disaster types [33]. Nevertheless, image-only approaches remain constrained by inconsistent image quality, limited contextual information, and dataset bias, which can hinder reliable severity estimation in complex real-world scenarios.

1.2.3. Multimodal-based approach

Recognizing the complementary nature of textual and visual information, recent research increasingly adopts multimodal fusion strategies to improve disaster classification and severity assessment (Table 3). Early multimodal studies demonstrated that combining text and image data improves disaster image retrieval and event classification compared with unimodal baselines [7,8,35]. Subsequent work applied CNN and Visual Geometry Group (VGG)-based visual encoders alongside recurrent or transformer-based text models, yielding consistent performance gains in disaster tweet classification and information extraction tasks [9,13]. More recent multimodal pipelines focus explicitly on damage and severity assessment. Gated multimodal systems trained on the CrisisMMD dataset learn adaptive modality weighting and report improved severity classification accuracy [10]. Cross-attention mechanisms further enhance robustness by modeling interactions between visual and textual cues, particularly for reducing misinformation in disaster-related content [11]. Integrating user-generated text and images has also proven effective for identifying informative disaster tweets under real-world conditions [13]. A Robustly Optimized BERT Pre-training Approach (RoBERTa)-VGG16 framework using multiplicative fusion demonstrates that jointly modeling both modalities outperform unimodal counterparts in disaster tweet classification [14].

Hybrid architectures bridging unimodal and multimodal paradigms also show promise. For example, ResNet50 combined with attention-based Long Short-Term Memory (LSTM) models improves disaster

Table 2
Image-based models for disaster classification and severity assessment.

Architecture	Year	Task Focus	Key Performance	Ref.
DenseNet201	2025	Flooded & decertified region detection	99.37% test accuracy	[25]
ResNet-50 / EfficientNet-B0	2024	Disaster image classification	96% average accuracy	[26]
CNN Ensemble + XGBoost	2023	Damage classification	95% accuracy, F1 = 0.96	[27]
SegFormer	2023	Semantic segmentation (UAV imagery)	mIoU: 96% (RescueNet), 82.22% (FloodNet)	[29]
YOLOv8	2024	Real-time damage detection (drone images)	High mAP and recall	[30]
ResNet50 + LSTM	2024	Image-text disaster classification	Significant performance improvement	[23]
GAN-based Augmentation	2023	Data augmentation for disaster images	+11% validation accuracy	[31]
VGG16 / VGG19	2024	Disaster image & video classification	97% (VGG19), 93% (VGG16)	[32]
DAVI	2024	Pseudo-label damage segmentation	Robust without ground truth	[33]
SLgViT	2024	Pixel-level building damage assessment	Accuracy 94.63% in Haidi dataset	[34]

Table 3
Summary of Studies on Multimodal Disaster Classification.

Study Focus	Methodology	Key Findings	Ref.
Context-aware multimodal fusion	Image-text fusion for disaster image retrieval	Improved disaster image retrieval performance	[7]
Social media disaster information mining	Joint text and image analysis	Effective extraction of disaster-related insights	[9]
Damage severity classification (CrisisMMD)	Gated multimodal fusion	Improved damage severity classification accuracy	[10]
False information reduction	Cross-attention-based multimodal model	Reduced misinformation in disaster content	[11]
Informative tweet identification	Text-image integration from user content	Reliable identification of informative disaster tweets	[13]
Multimodal tweet classification	RoBERTa (text) + VGG-16 (image), multiplicative fusion	Enhanced tweet classification performance	[14]

event classification and damage severity estimation by leveraging complementary representations [23][9]. Overall, multimodal systems consistently outperform unimodal approaches, particularly for severity-focused tasks where textual descriptions and visual evidence provide complementary information.

However, the application of vision-language pretraining models such as CLIP to disaster severity classification remains relatively limited [36]. While CLIP demonstrates strong zero-shot and few-shot capabilities in general vision-language benchmarks, its integration into disaster-specific multimodal pipelines-particularly for severity assessment-remains an emerging research direction rather than a standard practice. Similarly, XAI techniques, including Grad-CAM and SHAP, have been applied to individual disaster classification models to provide post-hoc visual or feature-level explanations [37–40]. Yet, their systematic incorporation into end-to-end multimodal frameworks for jointly interpreting text and image-based severity predictions is less commonly reported in the literature. Finally, semi-supervised and weakly supervised learning approaches have been widely explored in general computer vision and multimodal learning to address labeled data scarcity [41]. In contrast, their application to multimodal disaster severity classification remains comparatively limited, despite the abundance of unlabeled crisis data and the high cost of expert annotation. This gap motivates further investigation into scalable, interpretable, and data-efficient multimodal disaster analysis frameworks.

2. Methodology

2.1. Pipeline overview

ReliefNet follows a modular, step-by-step pipeline designed for scalable and interpretable disaster severity assessment. The pipeline consists of six stages: (1) data acquisition and preprocessing, (2) pseudo-label generation for unlabeled data, (3) ROI-based visual refinement, (4) unimodal learning for text and images, (5) accuracy-weighted multimodal fusion, and (6) post-hoc explainability. Each module operates independently and is connected through well-defined inputs and outputs, as illustrated in Fig. 1.

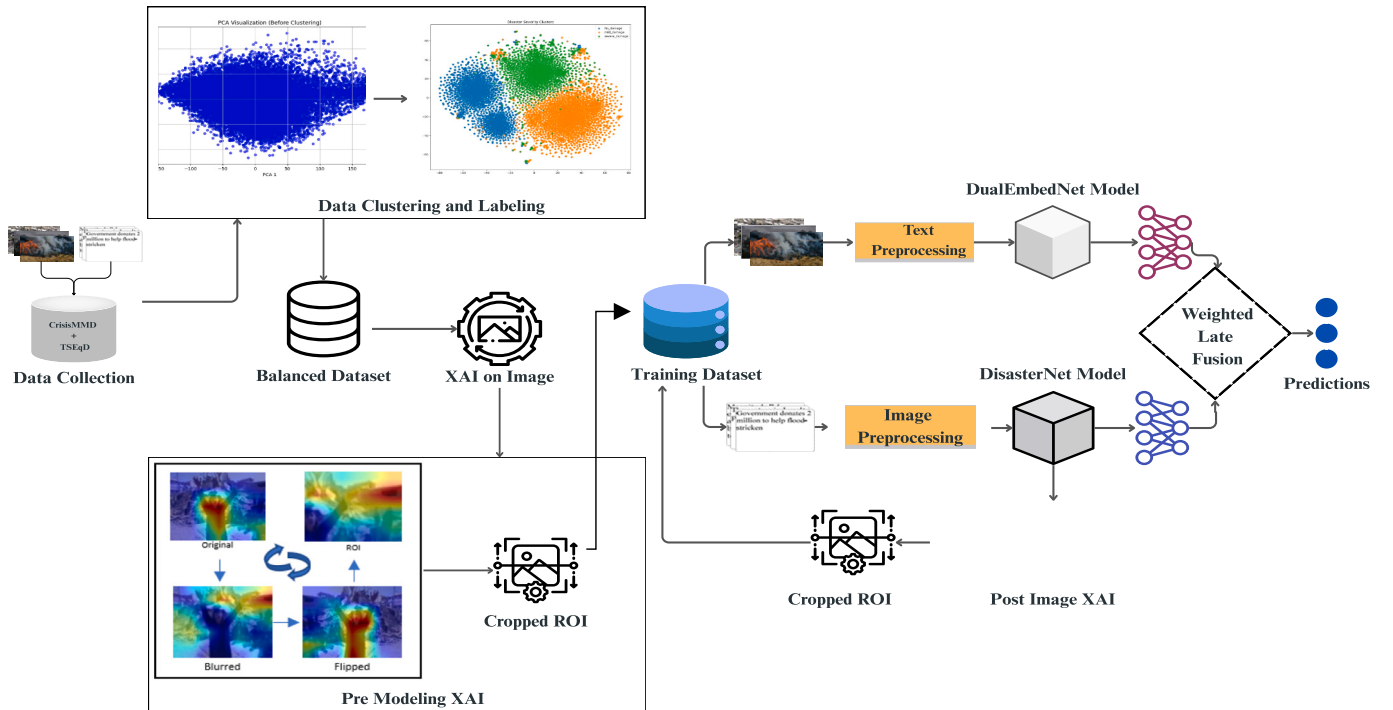


Fig. 1. Overview of the Methodology: A Detailed Flow of the Image and Text Feature Extraction, Clustering, and Classification Process.

2.2. Dataset acquisition and preprocessing

A core contribution of this work is the construction of a diverse and nearly balanced multimodal dataset for disaster severity prediction by integrating two complementary sources: CrisisMMD and the Turkey-Syria Earthquake Dataset (TSEqD) [1]. Both datasets contain aligned image-text pairs but differ substantially in label availability, class balance, and scene diversity. CrisisMMD provides 3526 image-text pairs annotated with three severity levels: little or no damage (475), mild damage (839), and severe damage (2212). Although these labels are high quality, the dataset is strongly imbalanced and limited in geographic and contextual diversity. In contrast, TSEqD contains 11,470 image-text pairs collected from a large-scale real-world disaster but lacks severity annotations. To leverage the complementary strengths of both datasets, we adopt a semi-supervised strategy in which TSEqD samples are pseudo-labeled and combined with CrisisMMD for supervised learning. Table 4 summarizes the dataset composition and preprocessing steps, while Fig. 2 and Fig. 3 illustrate the effect of pseudo-labeling on the TSEqD feature space.

2.3. Pseudo-Label generation for TSEqD

2.3.1. Multimodal feature extraction

For each image-text pair in (I_i, T_i) TSEqD, we extracted complementary unimodal and joint embeddings to capture both modality-specific and cross-modal semantics. Visual features were extracted using a pretrained ResNet50 backbone, mapping each image to a 2048-dimensional embedding:

$$v_i = f_{\text{ResNet}}(I_i), \quad v_i \in \mathbb{R}^{2048} \quad (1)$$

Textual features were obtained from a pretrained BERT-base model by extracting the [CLS] token representation:

$$t_i = f_{\text{BERT}}(T_i)[\text{CLS}], \quad t_i \in \mathbb{R}^{768} \quad (2)$$

To capture cross-modal alignment, we additionally extracted joint embeddings using CLIP, which projects both image and text into a shared semantic space:

Table 4
Summary of Dataset Composition and Processing Steps.

Dataset	Samples	Modalities	Label Status	Class Distribution	Processing Applied
CrisisMMD	3526	Image + Text	Available	Little/No: 475 Mild: 839 Severe: 2212	Labels used directly
TSEqD	11,470	Image + Text	Not available	Pseudo-labeled (k = 3)	ResNet (image features), BERT (text features), K-Means, Grad-CAM filtering, noisy sample removal
Final Dataset	14,996	Image + Text	Available	Little/No: 4748Mild: 5863Severe: 4379	Grad-CAM applied during preprocessing and training for semantic relevance and noise reduction

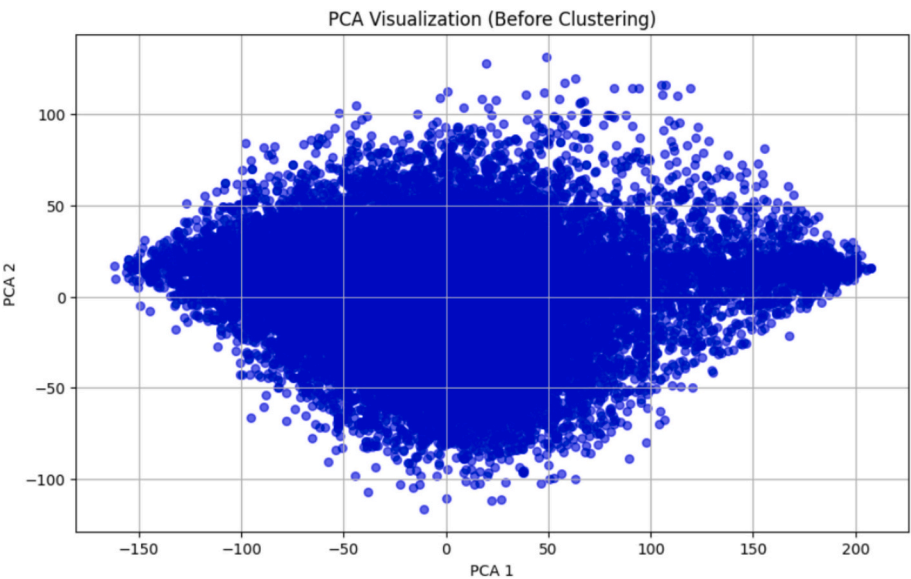


Fig. 2. Before clustering TSEqD Dataset.

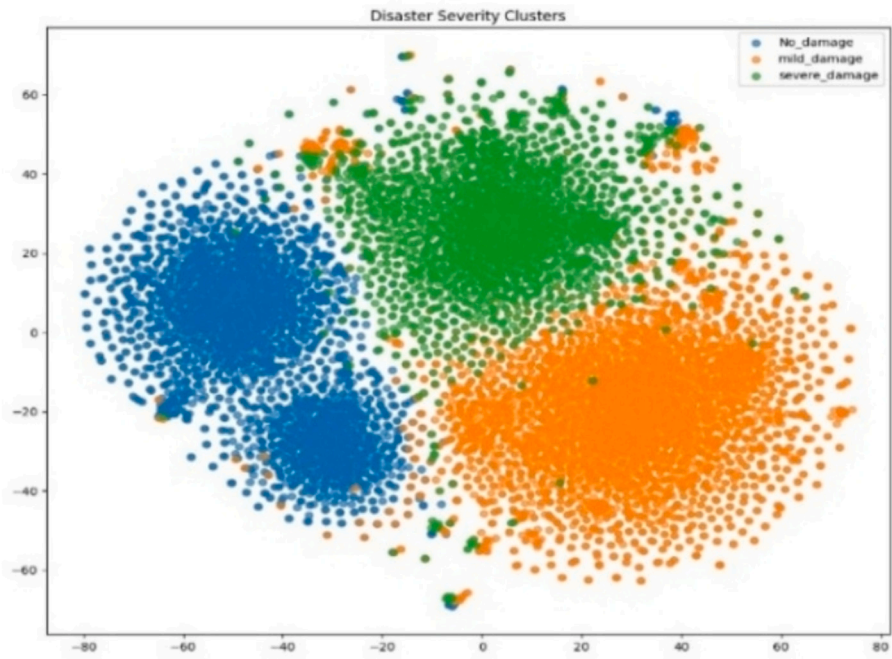


Fig. 3. After clustering TSEqD Dataset.

$$c_i = f_{\text{CLIP}}(I_i, T_i), \quad c_i \in R^{512} \quad (3)$$

The final multimodal representation was constructed by concatenating all embeddings:

$$x_i = [v_i; t_i; c_i] \in R^{2048+768+512} = R^{3328} \quad (4)$$

2.3.2. K-Means clustering and label assignment

We applied K-Means clustering to the multimodal embeddings x_i to generate pseudo-labels for TSeqD. The number of clusters was fixed to $k = 3$, matching the three severity categories defined in CrisisMMD. This choice was validated using the Silhouette Score to ensure cluster separability and semantic coherence. The clustering objective minimizes within-cluster variance:

$$\underset{\{\mu_j\}_{j=1}^k}{\operatorname{argmin}} \sum_{i=1}^N \min_j \|x_i - \mu_j\|^2 \quad (5)$$

Where, $\mu_j \in \mathbb{R}^{2816}$ represents the centroid of the j -th cluster. Each TSeqD sample was assigned a cluster label: $y_i \in \{0, 1, 2\}$, corresponding to little or no damage, mild damage, or severe damage. Cluster identities were aligned with CrisisMMD severity categories based on centroid similarity and class distribution. Fig. 2 shows the PCA visualization of TSeqD embeddings before clustering, while Fig. 3 demonstrates the improved separability after clustering.

2.3.3. Pseudo-label validation and data leakage prevention

To assess pseudo-label quality, a random subset of TSeqD samples was manually reviewed by human annotators and compared against expert severity judgments. Table 5 reports class-wise and overall agreement, with an overall accuracy of 90.3%. Severe and little/no damage classes exhibit particularly high agreement due to the presence of strong visual and textual cues, whereas mild damage shows lower agreement due to its inherently ambiguous nature. Cluster-label consistency was further evaluated using cluster purity (0.88) and Normalized Mutual Information (NMI = 0.84), as reported in Table 5. These results confirm that the pseudo-labels are sufficiently reliable for training. To prevent data leakage, pseudo-labeled TSeqD samples were used exclusively for training. All evaluation was conducted using strictly human-labeled data, either from the CrisisMMD test split or a separately annotated subset of TSeqD.

2.3.4. Dataset split description

After pseudo-labeling and clustering, the final dataset contains 14,996 image-text pairs across three severity classes. The dataset was split at the sample level into training, validation, and test sets using an approximately 80–10–10 ratio (Table 6), while preserving class proportions. ROI-refined and fully processed images were used only for training. Validation and test sets were kept strictly separate and unmodified to ensure unbiased evaluation. All hyperparameter tuning was performed on the validation set, and final results were reported exclusively on the test set.

2.3.5. Text pre-processing

Tweet text was preprocessed to reduce noise while preserving semantic content. Each raw sequence $T = \{t_1, t_2, t_3, \dots, t_n\}$ was normal-

Table 6

Dataset split and class distribution.

Dataset Split	Total Samples	Little / No Damage	Mild Damage	Severe Damage	Usage
Training (~80%)	11,991	3798	4690	3503	Model training (ROI-refined)
Validation (~10%)	1499	475	586	438	Model selection
Test (~10%)	1500	475	587	438	Final evaluation
Total	14,996	4748	5863	4379	–

Minor differences from exact 80–10–10 proportions arise from integer rounding while preserving exact class totals.

ized by converting tokens to lowercase, replacing URLs with a generic URL token, mapping user mentions to @USER, removing hashtag symbols while retaining keywords, and stripping punctuation, numerals, and extra whitespace. The cleaned text was tokenized into a sequence $X = (x_1, x_2, x_3, \dots, x_n)$ and mapped to integer indices based on a fixed vocabulary V , producing $S = (s_1, s_2, s_3, \dots, s_n)$, $s_i \in V(x_i)$. Sequences were padded or truncated to a fixed length l , yielding $S^l \in N^l$. Each token ID was then mapped to a dense embedding using an embedding matrix $E \in \mathbb{R}^{|V| \times d}$, resulting in the input tensor $H_0 \in \mathbb{R}^{l \times d}$ for downstream text models.

2.3.6. Image pre-processing and ROI refinement

To reduce background noise and align model attention with disaster-relevant regions, we applied ROI-based image refinement using Grad-CAM. Activation maps were generated using two pretrained CNNs (ResNet50 and VGG16) to ensure robustness and cross-architecture consistency. Grad-CAM heatmaps were computed for original and augmented images (including rotation, flipping, blurring, and contrast adjustment). Images where attention focused on semantically irrelevant regions were cropped to retain only the most salient disaster-related areas, such as collapsed structures or flooded zones. The refined images replaced the original samples in the training dataset.

This preprocessing step improves visual feature quality and stabilizes training by suppressing misleading background content, at the cost of modest preprocessing overhead.

2.3.7. Image post-processing and interpretability feedback loop

During post-training analysis, Grad-CAM, Grad-CAM++, and LIME were employed to interpret predictions from the DisasterNet image classifier. Class activation maps were used to identify failure cases where the model relied on irrelevant regions. Misclassified samples were manually inspected, and ROI cropping was reapplied when appropriate. Corrected samples were reinserted into the dataset with updated identifiers to preserve traceability.

This iterative loop of interpretation, manual refinement, and retraining improved robustness while maintaining transparency. Importantly, despite the large and diverse dataset, no evidence of overfitting was observed, confirming the stability of the proposed pre-processing and refinement strategy.

2.4. Evaluation metrics

Three classification settings were evaluated for each task: text-only, image-only, and multimodal (image + text). Model performance was assessed using accuracy, precision, recall, and F1-score, following [9]. All metrics were computed using a weighted-average strategy to account for class imbalance.

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN} \quad (6)$$

Table 5

Human audit and cluster consistency evaluation of Pseudo-labels.

Evaluation Type	Class / Metric	Value
Human Audit	Severe Damage Agreement (%)	93.6
	Mild Damage Agreement (%)	82.4
	Little / No Damage Agreement (%)	95.1
	Overall Agreement (%)	90.3
Cluster Metrics	Cluster Purity	0.88
	Normalized Mutual Information (NMI)	0.84

$$\text{Precision} = \sum_{i=1}^N \frac{TP_i}{TP_i + FP_i} \cdot \frac{\text{Num}_i}{\text{ALL}} \quad (7)$$

$$\text{Recall} = \sum_{i=1}^N \frac{TP_i}{TP_i + FN_i} \cdot \frac{\text{Num}_i}{\text{ALL}} \quad (8)$$

$$F1 - \text{score} = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \quad (9)$$

Accuracy measures the proportion of correctly classified samples. Precision reflects the correctness of positive predictions, while recall indicates the ability to identify actual positive samples. Here, TP, TN, FP, and FN stand for true positive, true negative, false positive, and false negative, respectively; subscript i , refers to the i -th class, N is the number of classes, Num_i is the sample count of class i , and ALL is the total number of samples. To ensure the reproducibility of results, we set the random seed to 42 for all experiments. Each experiment was repeated 3 times to assess the statistical reliability of the results and the consistency of the model's performance across different runs.

2.5. DualEmbedNet: text severity encoder

DualEmbedNet (Fig. 4.) serves as the textual severity inference module in the ReliefNet pipeline. It extracts complementary linguistic representations from disaster-related text and outputs class probabilities used later in multimodal fusion.

2.5.1. Input

Given a tokenized input sequence $X = (x_1, x_2, x_3, \dots, x_n)$ DualEmbedNet constructs an initial embedding matrix $H_0 \in \mathbb{R}^{n \times d}$ by summing three components:

$$H_0 = E_w(X) + E_p + E_s \quad (10)$$

Where, $E_w(X) \in \mathbb{R}^{n \times d}$ denotes the lexical embedding, $E_p \in \mathbb{R}^{n \times d}$ the positional embedding, $E_s \in \mathbb{R}^{n \times d}$ the segment embedding. Each embedding output is subjected to layer normalization and a dropout rate of 0.3 applied for regularization.:

2.6. Dual-encoder architecture

DualEmbedNet employs two parallel, heterogeneous transformer encoders, each consisting of 12 stacked layers. The dual-encoder design enables the model to capture complementary linguistic patterns and reduce representation bias associated with a single encoder. Each transformer layer $l \in \{1, \dots, L\}$ comprises:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V \quad (11)$$

Where, $Q = H_{l-1}W^Q$, $K = H_{l-1}W^K$, $V = H_{l-1}W^V$, and W^V , W^K , and $W^Q \in \mathbb{R}^{d \times d}$ are learnable parameters.

$$\text{FFN}(x) = W_2 \cdot \text{GELU}(W_1 x + b_1) + b_2 \quad (12)$$

Residual connections and layer normalization are applied after each sublayer to stabilize training. After the final transformer layer, a fixed-length sentence representation is extracted from the [CLS] token:

$$h = \tanh\left(W_{\text{pool}}H_L^{[0]} + b_{\text{pool}}\right) \quad (13)$$

The pooled outputs from both encoders, $h^{(1)}$, and $h^{(2)}$, are concatenated $h_{\text{fused}} = [h^{(1)}; h^{(2)}] \in \mathbb{R}^{2d}$. This fusion captures complementary linguistic perspectives and forms the input to the classification head. The fused representation is passed through a two-layer feedforward network:

$$z = \text{ReLU}\left(W_1^{\text{cls}} h_{\text{fused}} + b_1^{\text{cls}}\right) \quad (14)$$

$$y = W_2^{\text{cls}} \cdot \text{Dropout}(z) + b_2^{\text{cls}} \quad (16)$$

The output $y \in \mathbb{R}^C$, $C = 3$ represents the logits for the severity classes: little or no damage, mild damage, and severe damage.

2.7. Loss function and optimization

DualEmbedNet is trained using categorical cross-entropy loss.

Table 7
DualEmbedNet hyperparameters.

Parameter	Value
Encoder type	Dual heterogeneous transformers
Number of encoder layers	12 per encoder
Hidden dimension ((d))	768
Dropout rate	0.3
Pooling method	[CLS] token + tanh
Classification head	2-layer MLP
Activation function	GELU (encoder), ReLU (classifier)
Loss function	Categorical cross-entropy
Optimizer	Adam
Learning rate	1×10^{-5}
Batch size	32
Learning rate schedule	Linear decay with warm-up
Early stopping	Validation loss

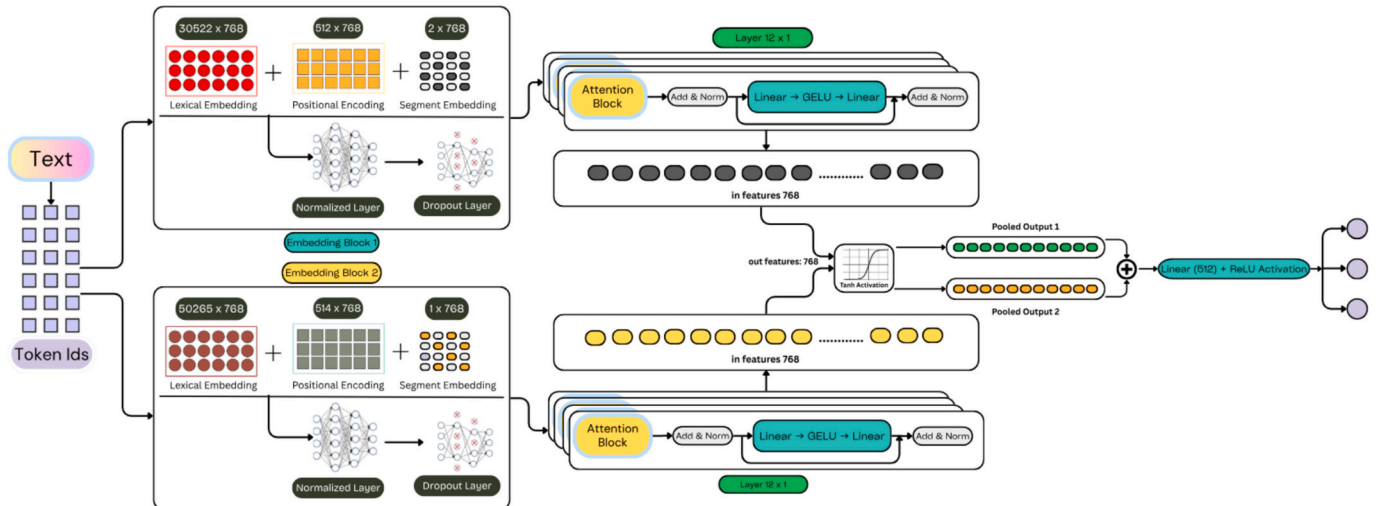


Fig. 4. The architecture of DualEmbedNet model.

Optimization is performed with the Adam optimizer, fine-tuning pre-trained encoder weights using a small learning rate to preserve linguistic knowledge while adapting to disaster-specific semantics. Early stopping is employed based on validation performance to prevent overfitting.

Algorithm 1: DualEmbedNet architecture.

```

1: for encoder ∈ {Encoder1, Encoder2} do
2:   Ew ← LexicalEmbedding(X) ∈ ℝn×d
3:   Ep ← PositionalEmbedding(n)
4:   Es ← SegmentEmbedding(n)
5:   H0 ← LayerNorm(Ew + Ep + Es)
6:   H0 ← Dropout(H0)
7:
8:   for ℓ = 1 to L do
9:     Q ← H(ℓ-1) WQ
10:    K ← H(ℓ-1) WK
11:    V ← H(ℓ-1) WV
12:    A ← softmax(QKT / √d) V
13:    A ← Dropout(A)
14:    Hℓ ← LayerNorm(H(ℓ-1) + A)
15:    FFN ← W2 · GELU(W1 Hℓ + b1) + b2
16:    Hℓ ← LayerNorm(Hℓ + FFN)
17:   end for
18:
19:   hencoder ← tanh(Wpool HL[0] + bpool)
20:   Store hencoder
21: end for
22:
23: hfused ← Concat(hencoder1, hencoder2)
24: z ← ReLU(W1cls hfused + b1cls)
25: z ← Dropout(z)
26: y ← W2cls z + b2cls
27:
28: return y
29: end function

```

DualEmbedNet provides robust textual severity representations by leveraging parallel transformer encoders and explicit feature fusion. This design (Algorithm 1) improves sensitivity to subtle linguistic cues commonly present in crisis-related text and supplies reliable text-based predictions for the multimodal fusion stage of ReliefNet. Table 7 explains the hyperparameters used during development.

2.8. DisasterNet model: visual severity representation module.

2.8.1. Input representation

DisasterNet (Fig. 5.) accepts an RGB image $X \in \mathbb{R}^{224 \times 224 \times 3}$ which is normalized and resized during preprocessing to ensure consistent spatial resolution across samples. This hierarchically extracts spatial and semantic features through five sequential convolutional blocks.

2.8.2. Convolutional backbone architecture

The network follows a five-block convolutional architecture that hierarchically extracts low-level to high-level visual features. Each block consists of two or three convolutional layers, where feature maps are computed as:

$$F^{(l)} = \text{ReLU}(W^{(l)} * F^{(l-1)} + b^{(l)}) \quad (17)$$

where $*$ denotes 2D convolution and $b^{(l)}$ is a bias term. Each block is followed by max pooling to reduce spatial resolution and improve translation invariance:

$$F_{\text{pooled}}^{(l)}(i, j, c) = \max_{(m, n) \in \Omega} F^{(l)}(i + m, j + n, c) \quad (18)$$

To model inter-channel dependencies and emphasize informative visual patterns, a channel-wise recalibration mechanism is applied after each pooling operation. First, global average pooling generates a channel descriptor:

$$z_c = \frac{1}{H \cdot W} \sum_{i=1}^H \sum_{j=1}^W F_c(i, j) \quad (19)$$

These descriptors are passed through a two-layer bottleneck network:

$$s = \sigma(W_2 \cdot \delta(W_1 \cdot z)) \quad (20)$$

Where, $\delta(\cdot)$ and $\sigma(\cdot)$ denote ReLU and sigmoid activations, respectively. The resulting scaling factors $s \in [0, 1]^C$ are applied channel-wise to recalibrate the feature maps. This mechanism enables DisasterNet to focus on visually salient disaster-related regions while suppressing irrelevant background information. After the final convolutional block, adaptive average pooling reduces the feature maps to a fixed spatial resolution of 1×1 , yielding a 512-dimensional feature vector: $v \in \mathbb{R}^{512}$. This representation is passed through two fully connected layers:

$$h_1 = \text{Dropout}(\text{ReLU}(W_1 \cdot v + b_1)), \quad W_1 \in \mathbb{R}^{4096 \times 512} \quad (21)$$

$$h_2 = \text{Dropout}(\text{ReLU}(W_2 \cdot h_1 + b_2)), \quad W_2 \in \mathbb{R}^{4096 \times 4096} \quad (22)$$

The final layer projects the features into the class space:

$$\text{output} = W_3 \cdot h_2 + b_3, \quad W_3 \in \mathbb{R}^{C \times 4096} \quad (23)$$

Class probabilities are obtained via softmax:

$$\hat{y}_i = \frac{e^{\text{output}_i}}{\sum_{j=1}^C e^{\text{output}_j}}, \quad i \in \{C_1, C_2, C_3\} \quad (24)$$

Where, C is the number of disaster severity classes, severe_damage, mild_damage, little_or_no_damage. The model has a total of 15.5 Giga Floating Point Operations Per Second (GFLOPs) (15.5 billion Floating-point Operations Per Second (FLOPs)). This indicates that the model strikes a balance between computational complexity and performance. The step-by-step model development is given in Algorithm 2, and the hyperparameters used (Table 8) during development.

Algorithm 2: DisasterNet image model.

```

1: for block = 1 to 5 do
2:   Apply 2-3 convolutional layers:
3:   F ← ReLU(Wconv * F + bconv)
4:   Apply max pooling:
5:   F ← MaxPool(F)
6:   Perform channel recalibration:
7:   for each channel c do
8:     zc ← (1 / H · W) ∑i ∑j Fc(i, j), ∀ c ∈ {1, ..., CF}
9:   end for
10:  s ← σ(W2 · ReLU(W1 · z))
11:  for each channel c do
12:    F̂c ← sc · Fc
13:  end for
14:  F ← F̂
15: end for
16: Apply adaptive average pooling to F:
17: v ← Flatten(AdaptiveAvgPool(F))
18: Pass through fully connected layers:
19: h1 ← Dropout(ReLU(W1 · v + b1))
20: h2 ← Dropout(ReLU(W2 · h1 + b2))
21: o ← W3 · h2 + b3
22: Compute softmax over output logits:

```

2.9. Multimodal fusion

As illustrated in Fig. 6, ReliefNet-Fusion integrates predictions from the image-based (DisasterNet) and text-based (DualEmbedNet) models to produce a final disaster severity decision. The module operates at the decision level, enabling adaptive weighting of modalities based on their predictive reliability.

2.9.1. Inputs

Image input X_{img} processed by DisasterNet, Text input X_{txt} processed

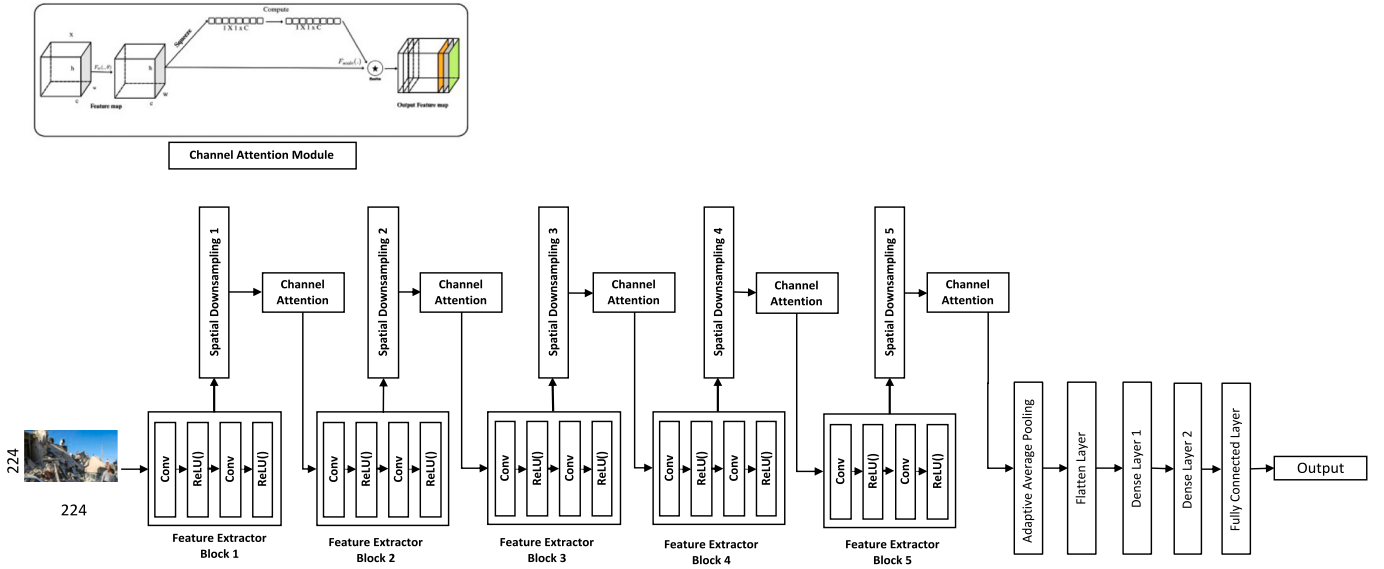


Fig. 5. The architecture of DisasterNet image model.

Table 8
DisasterNet hyperparameters.

Parameter	Value
Input resolution	(224 × 224 × 3)
Convolutional blocks	5
Channel recalibration	SE-style attention
Final feature dimension	512
Fully connected layers	4096 → 4096
Activation function	ReLU
Dropout rate	0.5
Loss function	Categorical cross-entropy
Optimizer	Adam
Learning rate	(1 × 10 ⁻⁴)
Batch size	32
Computational cost	15.5 GFLOPs

by DualEmbedNet. Each modality produces a class-probability distribution over C severity classes:

$$p^{(\text{img})} = \text{softmax}(f_{\text{img}}(x_{\text{img}})) \in R^C, \quad p^{(\text{txt})} = \text{softmax}(f_{\text{txt}}(x_{\text{txt}})) \in R^C \quad (25)$$

Where, f_{img} and f_{txt} denote the pretrained unimodal classifiers.

2.9.2. Fusion strategy

ReliefNet employs an accuracy-weighted decision-level fusion, where each modality contributes proportionally to its validation accuracy. Let a_{img} and a_{txt} denote the accuracies of the image and text models, respectively. The fusion weights are defined as:

$$w_{\text{img}} = \frac{a_{\text{img}}}{a_{\text{img}} + a_{\text{txt}}}, \quad w_{\text{txt}} = \frac{a_{\text{txt}}}{a_{\text{img}} + a_{\text{txt}}}, \quad w_{\text{img}} + w_{\text{txt}} = 1 \quad (26)$$

For example, with $a_{\text{img}} = 0.90$, and $a_{\text{txt}} = 0.95$, the resulting weights are $w_{\text{img}} = 0.4865$, and $w_{\text{txt}} = 0.5135$.

2.9.3. Fused prediction

The final fused probability distribution is computed as a convex combination of unimodal predictions:

$$p^{\text{fused}} = w_{\text{img}} p_{\text{img}} + w_{\text{txt}} p_{\text{txt}} \quad (27)$$

Class-wise probabilities are given by:

$$p_c^{(\text{fused})} = w_{\text{img}} p_c^{(\text{img})} + w_{\text{txt}} p_c^{(\text{txt})}, \quad \forall c \in \{1, 2, \dots, C\} \quad (28)$$

The final prediction is selected as:

$$\hat{y} = \arg \max p_c^{(\text{fused})} \quad (29)$$

2.9.4. Probabilistic interpretation

From a probabilistic perspective, this fusion strategy is equivalent to Bayesian model averaging, where each modality provides a conditional posterior distribution:

$$P(y = cx) \approx \sum_{m \in \{\text{img}, \text{txt}\}} w_m \cdot P_m(y = cx_m) \quad (30)$$

This formulation allows the model to marginalize over modality-specific uncertainties while emphasizing more reliable sources.

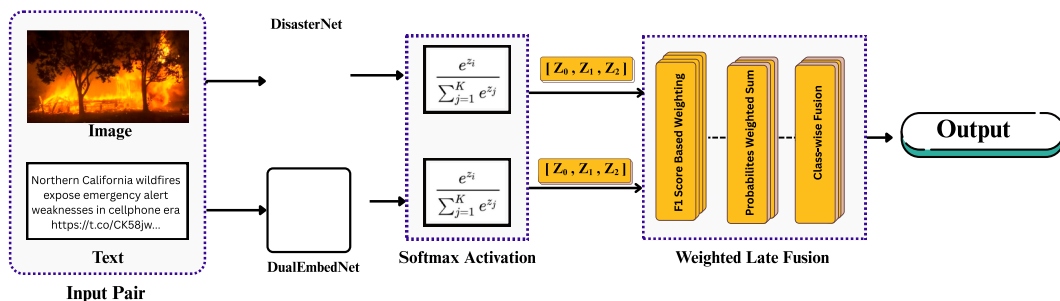


Fig. 6. Multimodal fusion process of our proposed architecture.

2.9.5. Computational characteristics

ReliefNet-Fusion is model-agnostic and training-free, requiring no additional optimization once the unimodal models are trained. It supports: asynchronous modality inference, graceful degradation when one modality is missing or unreliable, low computational overhead suitable for real-time deployment.

Algorithm 3: Multimodal weighted fusion.

```

1: function MultimodalFusion ( $x_{img}$ ,  $x_{txt}$ ,  $f_{img}$ ,  $f_{txt}$ ,  $\alpha_{img}$ ,  $\alpha_{txt}$ )
2:   # Step 1: Get softmax probability vectors from each modality
3:    $p_{img} \leftarrow \text{softmax}(f_{img}(x_{img}))$ 
4:    $p_{txt} \leftarrow \text{softmax}(f_{txt}(x_{txt}))$ 
5:   # Step 2: Normalize accuracies to compute modality weights
6:    $w_{img} \leftarrow \alpha_{img} / (\alpha_{img} + \alpha_{txt})$ 
7:    $w_{txt} \leftarrow \alpha_{txt} / (\alpha_{img} + \alpha_{txt})$ 
8:   # Step 3: Compute weighted fused probability vector
9:    $p_{fused} \leftarrow w_{img} * p_{img} + w_{txt} * p_{txt}$ 
10:  # Step 4: Select the class with the highest fused probability
11:   $y_{hat} \leftarrow \text{argc} \in \{1, \dots, C\} \text{ max } p_{fused}[c]$ 
12:  return  $y_{hat}$ 
13: end function

```

ReliefNet-Fusion (Algorithm 3 and Table 9) combines unimodal predictions through adaptive accuracy-weighted decision fusion, balancing robustness, interpretability, and efficiency. By dynamically adjusting modality contributions, the fusion module mitigates unreliable inputs and enables consistent performance across heterogeneous disaster data sources.

2.10. Explainable AI

XAI aims to make black-box machine learning models more transparent and trustworthy [42]. It uses techniques like counterfactuals, LIME, Grad-CAM, Grad-CAM++, and SHAP to clarify model decisions and ensure fairness. A graphical abstract of the process is shown in Fig. 7.

In this section, we explore DisasterNet Image model interpretability for disaster severity classification using three XAI techniques: Grad-CAM, Grad-CAM++, and LIME.

SHAP analysis indicates that the text-based component of ReliefNet prioritizes disaster-related keywords and tokens during prediction (Figs. 10–11), contributing significantly to the model's high precision and accuracy in distinguishing disaster severity levels. For the image-based component (Figs. 8–9), Grad-CAM, Grad-CAM++, and LIME were employed solely for post-hoc interpretability, error analysis, and controlled training refinement. These methods were not applied to evaluation data to avoid data leakage. Grad-CAM produced compact and semantically meaningful activation maps aligned with human-interpretable disaster cues, while Grad-CAM++ exhibited more diffused attention in complex scenes. LIME provided complementary super pixel-level explanations, aiding the diagnosis of errors caused by visually dominant but semantically irrelevant regions. ROI-guided refinement was applied only during training using Grad-CAM-based localization, while all dataset configurations and allocations are reported in Table 7. Attribution reliability was further validated using remove-and-measure sanity tests, where masking Grad-CAM-identified

regions led to the largest confidence reduction (−27%), followed by smaller but consistent drops for Grad-CAM++ and LIME. Failure cases were primarily observed under challenging visual conditions such as glare, smoke, reflections, and low-contrast debris.

Overall, Grad-CAM demonstrated stable and meaningful spatial localization suitable for constrained training refinement, while LIME remained effective for instance-level error analysis. Importantly, these explainability components were applied as bounded, non-iterative enhancements and did not influence evaluation results.

2.11. Data compliance and validation

The data acquisition pipeline (Fig. 12) was designed to ensure ethical, legal, and technical robustness. All web content was collected exclusively from the publicly accessible Al Jazeera News Media Gallery, which permits viewing for research and non-commercial academic purposes. Prior to scraping, the Terms of Service (ToS) were reviewed in detail, and only openly available images with accompanying captions were included. No user-generated or login-restricted content was accessed. Since all collected samples were news-agency photographs, the images do not contain personally identifiable information requiring consent, and each item retains its original copyright attribution as provided by the publisher. Metadata fields including photographer credit, publication date, and licensing statements were preserved. A controlled browser-automation setup (ChromeDriver) was used to render dynamic pages, enabling deterministic scrolling and loading of media elements. To ensure respectful usage, the scraper implemented rate-limiting, exponential backoff, and automatic error recovery, preventing excessive server load. Each image-caption pair was downloaded with a standardized naming convention, and corresponding metadata were stored in UTF-8 encoded tabular form to maintain multilingual compatibility.

Data collection employed a controlled browser-automation framework (ChromeDriver) to reliably render dynamic content. The scraper incorporated rate limiting, exponential backoff, and error recovery to prevent excessive server load. Image-caption pairs were stored using standardized naming conventions, with metadata saved in UTF-8 encoded tabular format. To enhance dataset quality, the following safeguards were applied:

1. **Deduplication:** Perceptual hashing (pHash) and Sentence-BERT cosine similarity were jointly used to identify and remove duplicate or near-duplicate image-caption pairs.
2. **Language Filtering:** fastText language identification was applied to retain only English-language disaster-related captions.
3. **Geographic and Event Diversity:** Metadata fields (event type, location, date) were used for stratified sampling to ensure coverage across multiple disaster types and regions.

To prevent methodological leakage, clustering was used only for preliminary severity grouping. Final labels were independently validated by two expert annotators, with disagreements resolved by a third expert following standardized guidelines. All evaluations therefore rely on expert-verified ground truth. ReliefNet Fusion showed strong agreement with these labels, demonstrating robustness across diverse disaster scenarios and media sources.

2.12. Optimization and inference efficiency

To assess real-time deployment feasibility, ReliefNet Fusion was optimized using a production-grade inference pipeline. The trained PyTorch model was exported to ONNX and compiled with NVIDIA TensorRT to leverage graph-level optimizations, mixed-precision execution, and hardware-aware scheduling. All experiments were conducted under fixed and reproducible conditions, with inference evaluated in a single-sample (batch size = 1) setting to reflect real-time

Table 9
ReliefNet-fusion parameters.

Parameter	Value
Fusion level	Decision level
Weighting criterion	Validation accuracy
Modalities	Image, Text
Number of classes (C)	3
Training required	No
Missing modality support	Yes
Computational overhead	Negligible

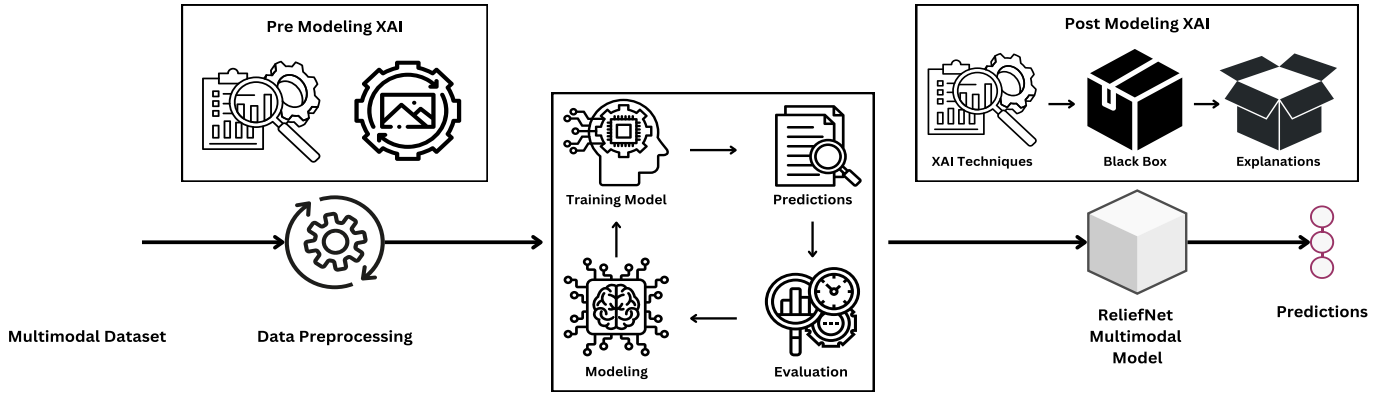


Fig. 7. Workflow of the explainable AI.

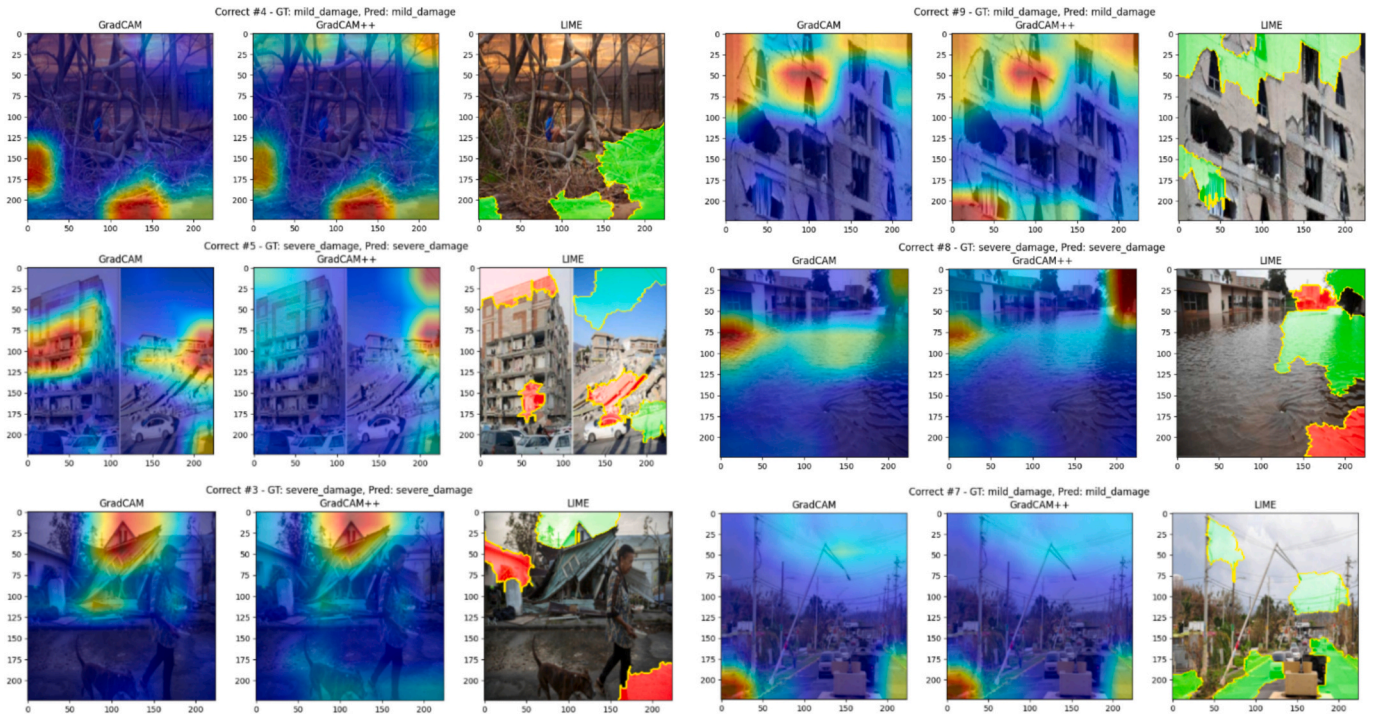


Fig. 8. XAI annotations on correct predictions.

operational constraints. Performance benchmarking focused on execution latency and throughput across FP32, FP16, and INT8 precision modes on both Graphics Processing Unit (GPU) and Central Processing Unit (CPU) platforms, enabling analysis of efficiency-accuracy trade-offs under deployment conditions. In parallel, deployment safeguards were integrated to support responsible use, including privacy-preserving local inference, removal of user-identifying information, and restriction of outputs to severity-level predictions suitable for decision support. A consolidated summary of system configuration, inference performance, optimization gains, and deployment safeguards is provided in Table 10.

2.13. Fuzzy logic and MCDM extensions

Given the inherent uncertainty in disaster severity assessment and the need for effective resource prioritization, we propose the integration of fuzzy logic and Multi-Criteria Decision-Making (MCDM) approaches. Fuzzy logic allows for the modeling of ambiguity in severity classification, offering a more nuanced representation of severity levels, such as between severe and mild damage (e.g., 97.66% F1 score for the image-

only model). MCDM techniques, such as Fuzzy Analytic Hierarchy Process (FAHP) and Technique for Order Preference by Similarity to Ideal Solution (TOPSIS), can prioritize disaster response by evaluating multiple criteria, including severity, accessibility (e.g., distance from emergency services), and reliability (e.g., infrastructure stability). These methods enhance the handling of conflicting priorities, improving resource allocation and intervention strategies, as demonstrated by our model's robust performance under missing-modality and modality-conflict scenarios, which were tested for graceful degradation in the absence of one modality or conflicting data.

3. Result and analysis

3.1. Overall performance

Table 11 summarizes the performance of text-only, image-only, and multimodal configurations across all evaluation metrics. Overall, all models achieve strong performance, reflecting the discriminative nature of disaster-related visual and textual cues. Among the unimodal

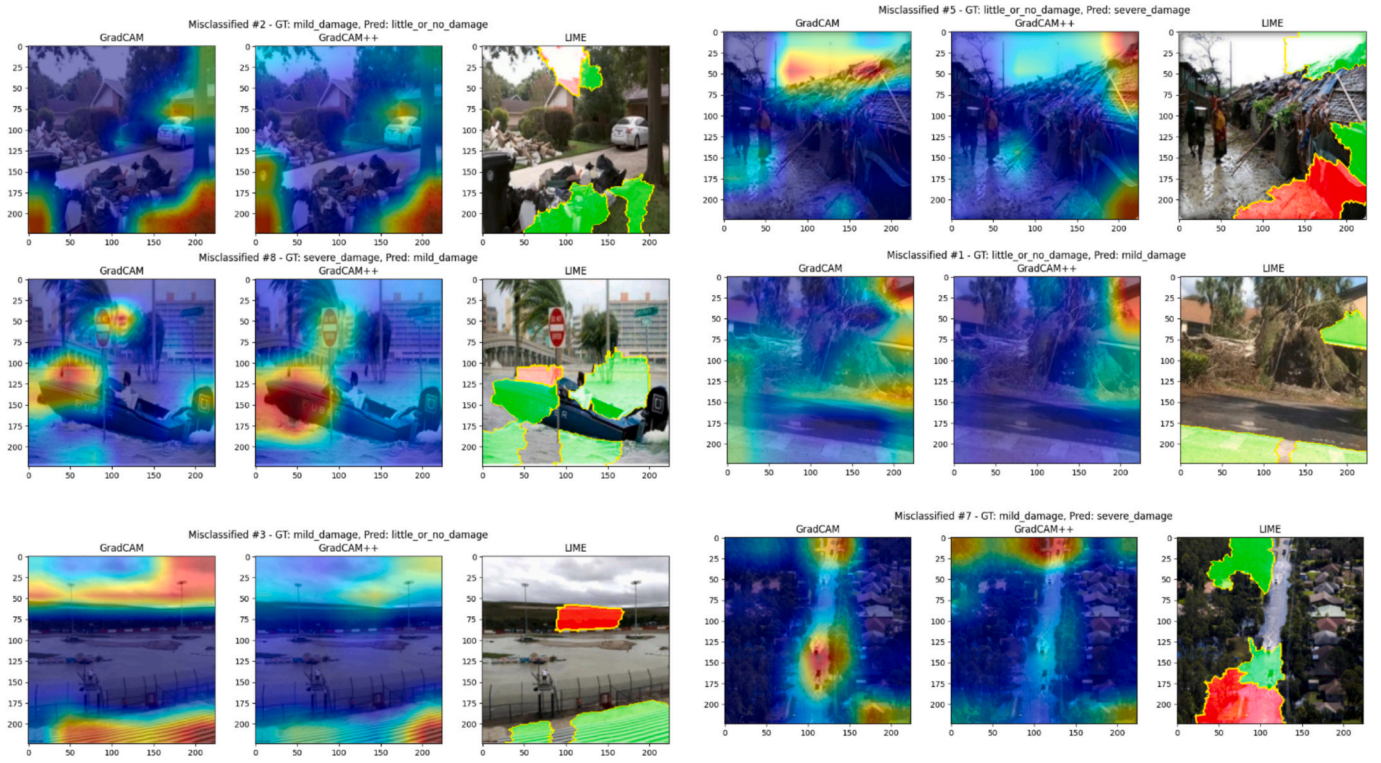


Fig. 9. XAI annotations on wrong predictions.

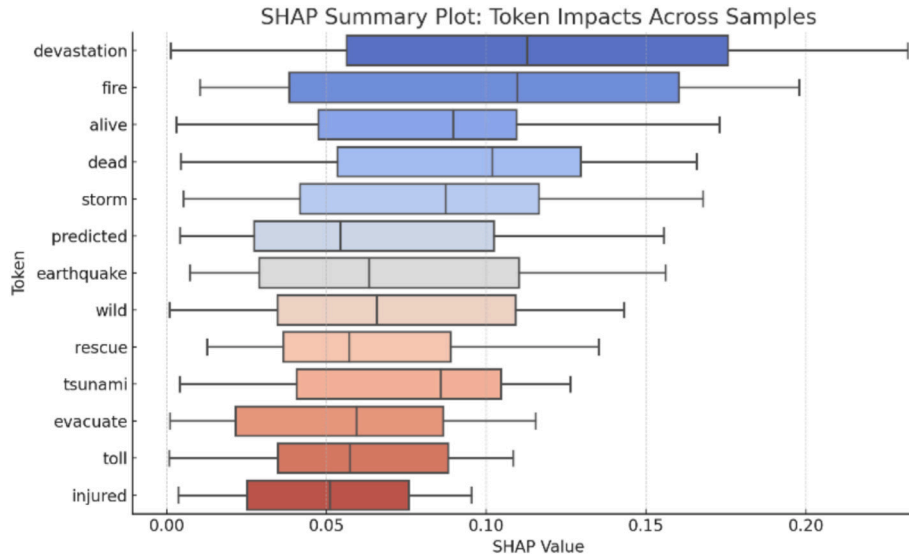


Fig. 10. Visualization token impacts across samples

approaches, the image-based model attains the highest aggregate F1-score (97.66%), while the proposed multimodal ReliefNet achieves comparable performance (97.00%) with improved robustness and interpretability.

While image-only inference yields the highest aggregate score, the margin over multimodal fusion is below 1% and is not statistically significant under paired resampling ($p > 0.05$). This observation highlights an important empirical characteristic of disaster datasets: visual evidence often dominates severity recognition, whereas textual content may introduce noise or ambiguity. Consequently, multimodal fusion does not always improve peak accuracy but instead contributes complementary strengths that become evident under challenging conditions.

3.2. Text and image encoder performance

To better understand modality behavior, we evaluate the text and image encoders independently. Tables 12 and 13 report class-wise performance for DualEmbedNet and DisasterNet, respectively.

The image encoder consistently outperforms the text encoder across all severity classes, reflecting the strong discriminative power of explicit visual damage cues such as collapsed infrastructure, flooding, and debris. In contrast, the text encoder performs particularly well for mild damage, where captions often provide contextual qualifiers not always evident visually. These complementary error profiles motivate multimodal integration.

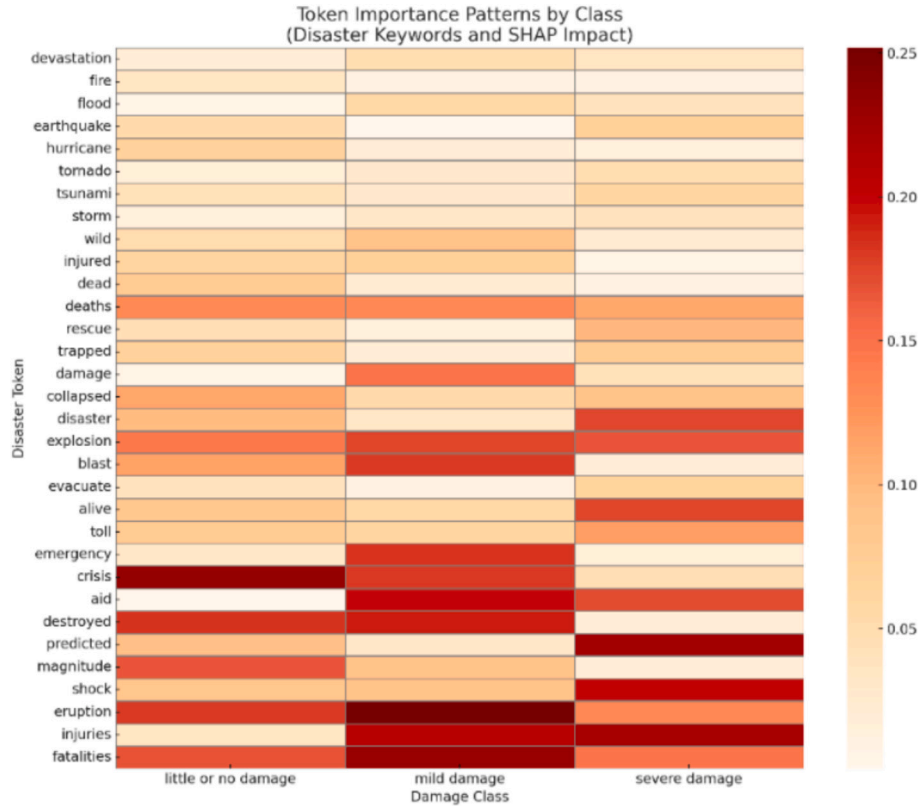


Fig. 11. Token importance patterns by class

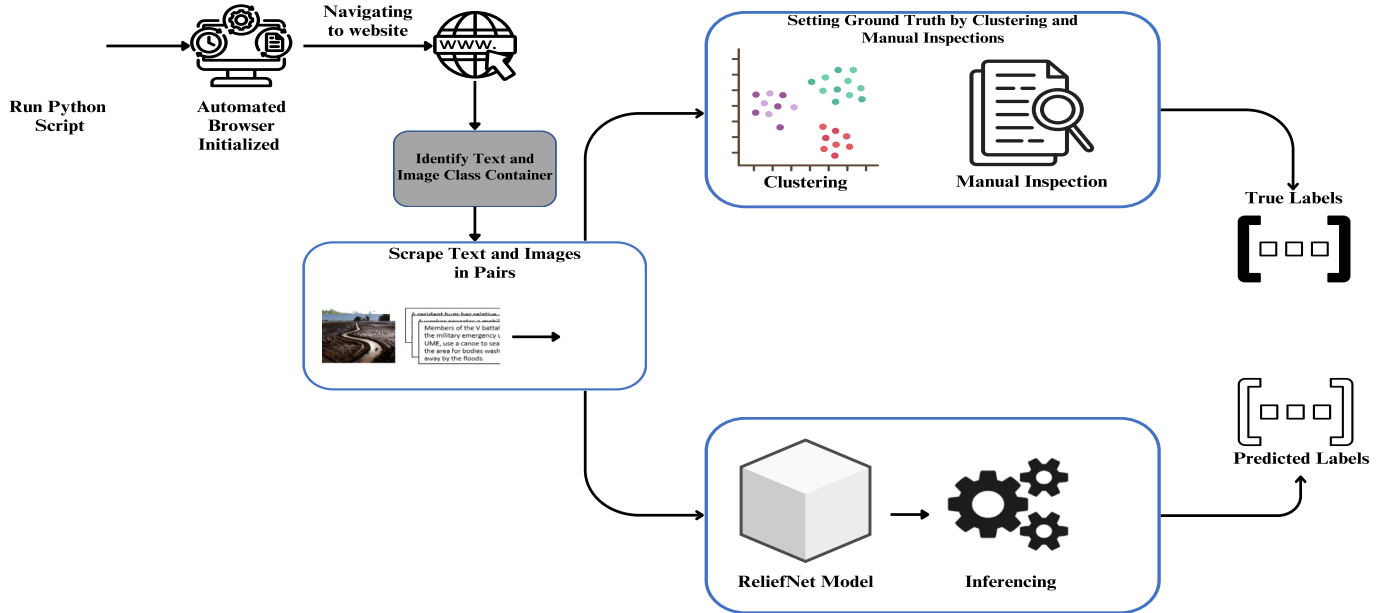


Fig. 12. Web scraping and validation pipeline.

3.3. ReliefNet multimodal performance

The proposed ReliefNet model integrates DualEmbedNet and DisasterNet through decision-level fusion. Its classification performance is summarized in Table 14.

Although the image-only model achieves marginally higher aggregate F1, multimodal fusion consistently reduces severe misclassification errors, particularly in cases where visual damage cues are subtle or

partially occluded. Conversely, multimodal fusion can underperform image-only inference when captions are off-topic, emotionally expressive without factual content, or reference prior events unrelated to the visible scene. This behavior explains the slight reduction in peak F1-score and underscores the importance of adaptive weighting rather than naive fusion.

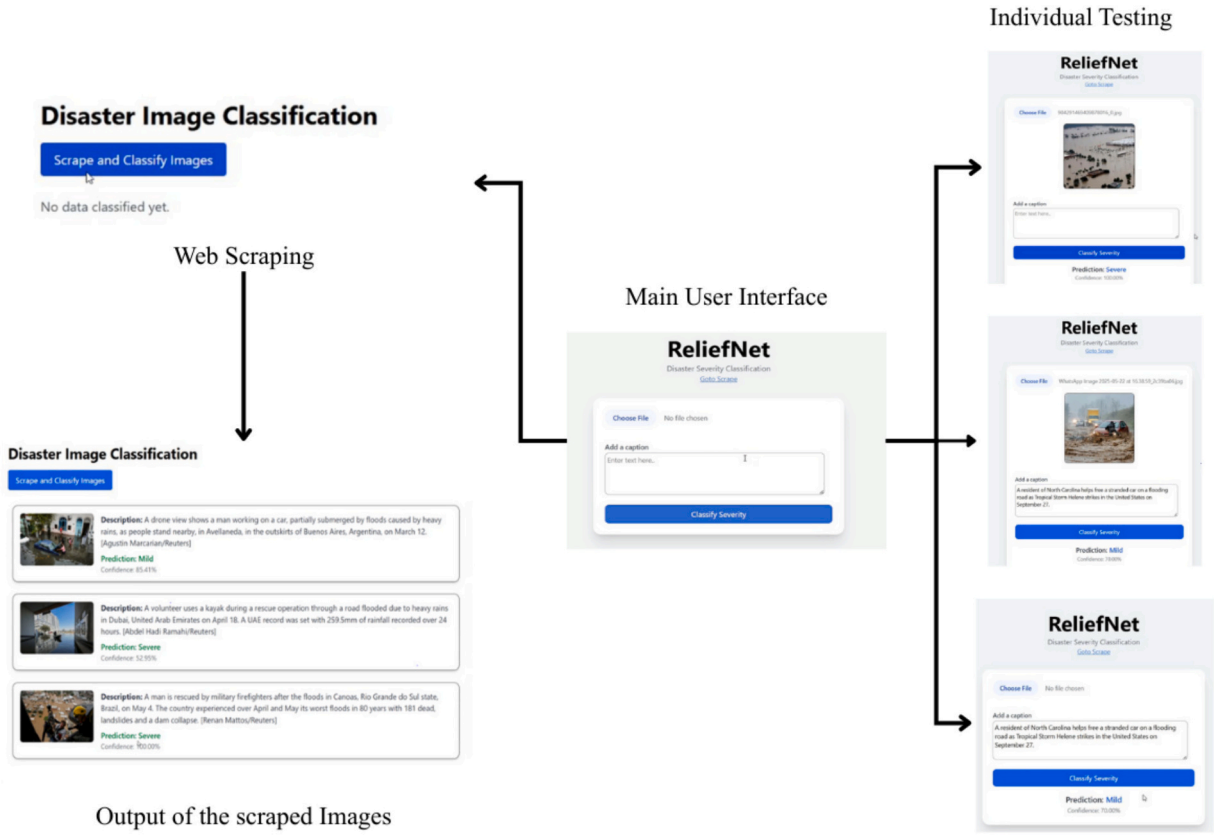


Fig. 13. Web-based platform for disaster severity classification.

3.4. Robustness to missing modality

Operational disaster streams frequently contain incomplete data. To evaluate robustness, we tested all models under missing-modality conditions (Table 15).

ReliefNet demonstrates graceful degradation, outperforming unimodal baselines even when one modality is absent. This property is critical for real-world deployment, where images or captions may be delayed, corrupted, or unavailable.

3.5. Fusion ablation study

To assess fusion behavior, we compare equal-weight fusion with accuracy-weighted fusion (Table 16).

Equal fusion degrades performance by over-amplifying noisy textual signals. In contrast, accuracy-weighted fusion dynamically emphasizes the more reliable modality while preserving complementary information, yielding consistent improvements over naive fusion and greater robustness across scenarios.

3.6. Stability and calibration

Across multiple runs, ReliefNet achieves stable F1-scores between 95% and 97%, with variability quantified using the standard deviation defined in Eq. (31). Expected Calibration Error (ECE) values of 0.0392 for DualEmbedNet and 0.0387 for DisasterNet (computed using 15 confidence bins) indicate well-calibrated probability estimates, supporting reliable decision-making in time-critical applications:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2} \quad (31)$$

where, N is the number of evaluation runs, x_i is the F1 score from the i^{th}

run, and μ is the mean F1 score across all runs.

3.7. Comparison with State-of-the-Art (SOTA) methods

Evaluating disaster classification models is complicated by differences in datasets, task definitions, and evaluation protocols across studies. To ensure a fair and transparent comparison, we structure our analysis along two axes:

1. Contextual comparison with published state-of-the-art methods, grouped by dataset and task alignment (Table 17).
2. Comparison under identical data splits using strong baselines (Table 18).

Table 17 summarizes comparisons with published methods, grouped by evaluation comparability. Studies using the same datasets and tasks but different splits are considered partially comparable, while those using different datasets or tasks are included for contextual reference only.

ReliefNet outperforms all baselines under identical conditions, improving multimodal F1 by 4.7 percentage points over BERT + ResNet fusion and by 18.6 points over CLIP zero-shot inference. The large gap with CLIP highlights the domain mismatch between generic vision-language pretraining and disaster-specific severity assessment, underscoring the benefit of task-aligned fine-tuning and fusion.

Severity Classification Performance: The closest prior study on severity assessment reports an F1 score of 67.0% on CrisisMMD images [43]. In contrast, DisasterNet achieves 97.66% F1 on the combined CrisisMMD + TSeqD dataset. Although differences in data splits limit direct comparability, the large performance margin indicates substantial gains from deeper visual encoders, ROI-focused preprocessing, and increased training diversity.

Effectiveness of Multimodal Fusion: Earlier multimodal disaster-classification approaches typically report F1 scores between 80% and 87% across related tasks [43]. ReliefNet improves this substantially,

Table 10

Deployment configuration, inference performance, optimization impact, and safeguards for ReliefNet.

Category	Aspect	Specification / Result
System Configuration	Hardware	NVIDIA RTX 3060 (12 GB VRAM), Intel i7 CPU
	Framework	PyTorch 1.14 → ONNX 1.14 → TensorRT
	Input Resolution	224 × 224
	Batch Size	1
	FP32 GPU Latency	32.4 ms
	FP32 Throughput	30.9 fps
	FP32 Latency	87.6 ms
	FP32 Accuracy	Baseline
	Impact	
	FP16 GPU Latency	18.7 ms
Inference Precision Performance	FP16 GPU Throughput	53.5 fps
	FP16 CPU Latency	61.2 ms
	FP16 Accuracy	None observed
	Impact	
	INT8 GPU Latency	10.3 ms
	INT8 GPU Throughput	96.8 fps
	INT8 CPU Latency	46.3 ms
	INT8 Accuracy	Negligible (<0.2%)
	Impact	
	Text Encoder	0.532 s → 0.019 s (28× speedup)
Module-wise Latency (GPU, INT8)	Image Encoder	0.174 s → 0.014 s (12× speedup)
	Fusion Module	0.576 s → 0.076 s (7.5× speedup)
	Privacy	Local inference only; no data transmitted or stored
Deployment Safeguards	User Identifiers	Removed prior to inference
	Text Handling	Tokenized in-memory; no raw text logging
	Image Handling	No image persistence after inference
	Misuse Prevention	Severity-only output (no personal or location inference)
	Human-in-the-loop	Outputs support decisions, not automated actions

Table 11

Performance summary across models.

Approach	Accuracy	Precision	F1-Score
Text-Based	94.50	95.00	95.00
Image-Based	97.66	97.00	97.66
Multimodal-Based	97.00	97.00	97.00

Table 12

DualEmbedNet classification performance.

Class	Precision	Recall	F1-score
Severe Damage	0.927 ± 0.020	0.927 ± 0.020	0.927 ± 0.020
Mild Damage	0.975 ± 0.015	0.975 ± 0.015	0.975 ± 0.015
Little / No Damage	0.939 ± 0.023	0.939 ± 0.023	0.939 ± 0.023
Overall Accuracy			0.945 ± 0.019

Table 13

DisasterNet classification performance.

Class	Precision	Recall	F1-score
Severe Damage	0.961 ± 0.018	0.961 ± 0.018	0.961 ± 0.018
Mild Damage	0.972 ± 0.014	0.972 ± 0.014	0.972 ± 0.014
Little / No Damage	0.965 ± 0.025	0.965 ± 0.025	0.965 ± 0.025
Overall Accuracy			0.968 ± 0.015

Table 14

ReliefNet classification performance.

Class	Precision	Recall	F1-score
Severe Damage	0.970 ± 0.021	0.970 ± 0.021	0.970 ± 0.021
Mild Damage	0.964 ± 0.018	0.964 ± 0.018	0.964 ± 0.018
Little / No Damage	0.955 ± 0.016	0.955 ± 0.016	0.955 ± 0.016
Overall Accuracy			0.963 ± 0.024

Table 15

Performance under missing-modality conditions.

Model Configuration	Precision (%)	Recall (%)	F1-score (%)
Text-only	93.8	93.8	93.8
Image-only	95.7	95.7	95.7
Multimodal (ReliefNet)	96.8	97.2	96.6

Table 16

Fusion strategy ablation results.

Fusion Strategy	Accuracy (%)	F1-score (%)
Text-only	94.5	95.0
Image-only	97.66	97.66
Equal Fusion	96.1	96.4
Accuracy-Weighted Fusion (Ours)	97.0	97.1

achieving 97.0% F1. This consistent gain across heterogeneous comparisons suggests that accuracy-weighted decision-level fusion effectively exploits complementary modality strengths while mitigating noise from weaker inputs.

Trade-offs in Open-World and Zero-Shot Settings: Open-set and zero-shot approaches prioritize robustness to unseen disaster categories but generally sacrifice accuracy, with reported F1 scores around 97.97% [46]. Our adapted open-world baseline improves performance to 82.1%, while the fully supervised ReliefNet reaches 97.0% F1. These results highlight a practical trade-off: open-world robustness versus high-accuracy severity estimation in operational deployments.

Unimodal Contextualization: Image-only models often achieve strong results on narrowly scoped tasks such as hazard or crack detection [49]. DisasterNet's 97.66% F1 is competitive with these specialized methods while addressing a more challenging three-class severity classification problem. On the text side, DualEmbedNet attains 95.0% F1, aligning closely with state-of-the-art text-based models and confirming the effectiveness of language cues for severity estimation.

3.8. Web-based platform

To demonstrate real-time applicability, we developed a web-based platform (Fig. 13) that allows users to upload disaster-related text and images while automatically displaying relevant, web-scraped content from credible news sources. The platform integrates our optimized multimodal models to perform real-time disaster severity classification, providing responsive inference, interactive visualization, and continuous monitoring of unfolding events. The access details, supported formats, latency characteristics, and throughput performance of the ReliefNet platform are summarized in Table 19.

3.9. External validation

External samples were collected from the Al Jazeera News Media Gallery, focusing on recent disaster events reported in early 2025, including Tropical Cyclone Gamane and flooding events in Iraq. These events introduce substantial domain shift relative to the training data in terms of geographic region, disaster type, visual composition, and caption style. A total of 73 image-text pairs were scraped using a compliant and rate-limited pipeline.

All samples were independently annotated by domain experts using

Table 17

Comparison with SOTA methods for disaster classification.

	Ref.	Dataset	Split (train/val/test)(%)	Task	Algorithm	Reported F1 (%)	Our F1 (%)	Comparability
Same Dataset (CrisisMMD and/or TSEqD)	[43]	Hurricane Harvey Tweets	Not specified	Urgent Help Detection	CNN + SVM + MLP	86.0	98.23	Partial †
	[44]	CrisisMMD + TSEqD	Not specified	Damage Assessment	ResNet-50 + BERT + IDEA	–	–	Partial §
	[11]	CrisisMMD	78.8/10.3/10.9	Damage Severity	SSE-Cross-BERT-DenseNet	70.41	98.23	Full §
	[45]	CrisisMMD	60/20/20	Multi-Modal Classification	Fine-tuned DenseNet + Fine-tuned BERT	~ (66–88)	98.23	Partial
	[46]	PyImage Search readers	60/17/23	Disaster Detection and Classification	Multilayered Deep CNN	–	–	No§
Different Dataset	[47]	CrisisNLP	78.8/10.3/10.9	Damaged Structure with Severity	Multitask DeiT ALBERT	~ (79–86)	98.23	No
	[48]	LEVIR-CD	69.5/10/20.1	Remote Sensing Change Detection	ChangeCLIP	~ (91.01–91.30)	–	No
	[36]	DisasterM3	–	Disaster Recognition, Assessment, Reporting	Fine-tuned: Qwen2.5-VL-7B, InternVL3-8B	–	–	No
	[37]	Natural Disaster Datasets and MEDIC	98.5/1.5	Disaster Images Classification	ViT-B-32	–	–	No
	[38]	xBD Dataset	74.0/–/26	Identifying Severity Post-Disaster	MA_ShallowNetV2, MA_ShallowNetV3	–	–	No

§ = Same Dataset, Same Task; ‡ = Same dataset, different task; † = Different Dataset, Same Task, * = Different Dataset, Different Task.

Table 18

Baseline comparison on identical data splits.

Model	Architecture	Text F1	Image F1	Multimodal
BERT + ResNet + Late Fusion	BERT [CLS] + ResNet50 + MLP	89.4	89.4	89.4
CLIP (Zero-Shot)	ViT-B/32 + Prompting	–	76.2	78.4
Open-World Baseline [46]	Multimodal Open-Set	–	–	82.1
ReliefNet (Ours)	DualEmbedNet + DisasterNet + Weighted Fusion	95.0	97.66	97.0

Table 19

Access details and usage instructions.

Platform URL	https://disasterseverity.netlify.app/
Demo Video URL	https://drive.google.com/file/d/1NB3iAfJRvainjyT_yD4z-9uiCyUi5e68/view?usp=drive_link
Access Requirement	publicly accessible and does not require authentication.
Supported Input Formats	Images: .jpg, .png, .jpeg (≤ 5 MB); Plain text captions up to 500 characters
Typical Latency	Image-only inference: ~0.8–1.2 s; Text-only inference: ~0.3–0.5 s; Multimodal inference: ~1.4–1.8 s
Data Throughput	In testing, ~20 requests/min without performance degradation on a single GPU backend.

the same three severity categories defined in this study. The resulting class distribution was: severe damage (39), mild damage (16), and little or no damage (18). These labels were used exclusively to establish ground truth for evaluation. The external dataset was strictly excluded from training, validation, pseudo-labeling, ROI refinement, and model selection, ensuring a leakage-free assessment.

Table 20 reports ReliefNet performance on the externally collected dataset. Despite the small sample size and pronounced domain shift, ReliefNet maintains stable performance, particularly for severe damage cases, which are most critical for humanitarian response.

These results demonstrate that ReliefNet generalizes effectively to unseen, real-world disaster scenarios encountered through live news streams, while exhibiting predictable degradation under ambiguous

visual evidence and sparse or indirect textual descriptions.

4. Discussion

This study introduced ReliefNet, a real-time multimodal framework for disaster severity assessment, and analyzed its behavior relative to prior approaches. Rather than reiterating performance figures, this discussion highlights the mechanisms that drive ReliefNet's behavior, its limitations, and the practical trade-offs relevant to deployment. ReliefNet's effectiveness is primarily driven by three design choices. First, CLIP-based vision–language representations provide strong cross-modal alignment, enabling the system to interpret noisy captions and visually complex scenes typical of real disaster reporting. Unlike earlier pipelines relying on isolated CNN or text embeddings, joint representations allow complementary cues across modalities to reinforce each other rather than compete. Second, accuracy-weighted decision-level fusion improves robustness by adapting to modality reliability at inference time. Visual evidence dominates when images clearly depict structural damage, while textual cues gain importance when images are ambiguous but captions are informative. This adaptive behavior explains why weighted fusion is more stable than equal-weight fusion and why multimodality does not always outperform image-only models in clean visual scenarios. Third, ROI-focused preprocessing improves visual reasoning by suppressing irrelevant background content. Error analysis showed that failures often arise from cluttered scenes, peripheral objects, or visually dominant but semantically irrelevant regions. Emphasizing disaster-relevant regions such as collapsed structures or flooded areas stabilizes predictions under real-world conditions.

ReliefNet generalizes across CrisisMMD and TSEqD due to its exposure to diverse geographic contexts, architectural styles, and reporting conventions. Multimodal alignment and semi-supervised labeling reduce sensitivity to annotation noise, which is common in severity assessment where class boundaries are subjective. Nevertheless,

Table 20External validation performance on web-scraped news data ($n = 73$).

Class	Precision (%)	Recall (%)	F1-score (%)
Severe Damage	91.8	94.9	93.3
Mild Damage	78.6	75.0	76.8
Little / No Damage	83.3	77.8	80.4
Overall (Weighted)	87.6	88.9	88.2

limitations remain. The model struggles with visually symbolic images lacking explicit damage cues, captions referencing prior events without visible impact, and borderline “mild damage” cases where both visual and textual signals are inherently ambiguous. These failures reflect fundamental constraints of inferring severity from indirect or incomplete evidence and motivate future incorporation of temporal context and external signals. From a deployment perspective, ReliefNet balances accuracy, latency, and interpretability. Optimized inference enables real-time operation, while decision-level fusion and post-hoc XAI provide transparency without imposing heavy computational overhead. This balance makes ReliefNet suitable for operational disaster-monitoring settings, where robustness and explainability are as critical as raw accuracy.

5. Conclusion, limitations, and future work

This paper presented ReliefNet, an explainable multimodal framework for disaster severity classification from social media image–text pairs. The study addressed three persistent challenges in this domain: noisy multimodal inputs, limited labeled data, and the need for interpretable, real-time decision support. ReliefNet integrates two complementary unimodal models: DualEmbedNet for textual severity inference and DisasterNet for visual damage assessment. These components are combined through an accuracy-weighted, decision-level fusion strategy that adapts to modality reliability at inference time. This design preserves strong unimodal performance while improving stability under missing or unreliable inputs. Experimental results on a unified dataset of 14,996 samples show that ReliefNet achieves competitive performance relative to strong image-only baselines, while offering improved robustness, calibration, and graceful degradation in incomplete-data scenarios.

Beyond model architecture, the work contributes a scalable data construction and validation pipeline. By merging CrisisMMD with the previously unlabeled TSEqD dataset, and introducing multimodal K-Means pseudo-labeling with human audit and cluster-quality validation, the study provides one of the largest structured datasets for disaster severity classification. The use of Grad-CAM-guided ROI refinement and post-hoc explainability methods further supports data quality assurance and model transparency without contaminating evaluation.

Finally, the system was designed with deployment constraints in mind. ReliefNet operates in real time on a single GPU, supports optimized inference across multiple precision modes, and incorporates practical safeguards for responsible use. Taken together, these contributions demonstrate that adaptive multimodal fusion, combined with careful data curation and explainability, can deliver reliable and interpretable disaster severity assessment suitable for operational humanitarian decision-making.

5.1. Limitations

1. Data limitations. Severity annotations for TSEqD rely partly on unsupervised pseudo-labeling, which may introduce residual noise despite manual checks. The dataset is limited to English-language content and focuses primarily on earthquakes, floods, and hurricanes, restricting coverage of other hazards and multilingual settings.
2. Methodological limitations. The current decision-level fusion does not model fine-grained cross-modal interactions. Severity is treated as a three-class problem, which may oversimplify the continuous and subjective nature of disaster impact. Model confidence is not explicitly calibrated.
3. Deployment limitations. Reported inference speedups depend on specific GPU hardware and may vary across platforms. External validation was conducted using a single news source, and the web platform has not yet undergone formal security auditing.

5.2. Future work

1. To address data limitations, future work will incorporate satellite imagery, government reports, and additional news feeds, and extend multilingual support using models such as mBERT or XLM-R. Broader disaster categories will be included to improve generalization.
2. To address methodological limitations, feature-level fusion with cross-modal attention will be explored, along with uncertainty-aware modeling and formal probability calibration. Continuous or ordinal severity modeling will also be investigated to better capture gradations of impact.
3. To address deployment limitations, future efforts will benchmark performance across heterogeneous hardware, integrate drift detection and retraining strategies, and conduct security and robustness audits for operational readiness. Human-in-the-loop workflows and fairness audits across regions and demographics will be strengthened prior to real-world deployment.

Data availability, reproducibility, and ethics

All training code, preprocessing pipelines, model checkpoints, optimization scripts, and evaluation protocols will be released upon publication to support reproducibility. Data access will comply with platform terms and ethical guidelines. ReliefNet does not retain personal identifiers, avoids facial recognition, and is intended strictly for decision support. Deployment is recommended only with human oversight, and licensing will prohibit surveillance, discrimination, or misuse.

CRediT authorship contribution statement

Khondokar Radwanur Rahman: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Amith Khandakar:** Formal analysis, Data curation, Conceptualization. **Md. Fahim Hossen:** Writing – original draft, Visualization, Methodology. **Shaikh Golam Rabbani:** Writing – original draft, Formal analysis. **Aniruddha Saha:** Writing – original draft, Data curation. **Md. Faysal Ahamed:** Project administration, Methodology. **Md. Munawar Hossain:** Methodology, Conceptualization. **Mohamed Arselene Ayari:** Methodology, Investigation. **Rajab Abdulla R.E. Al-Esmail:** Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work was supported by the project MCSC02-0301-250022, titled “Leveraging AI and Blockchain for Enhanced Disaster Response and Transparent Donation Management in the Charitable Sector.” The authors would also like to acknowledge the use of AI tools, including ChatGPT and QuillBot, for assistance in improving the clarity and presentation of the manuscript.

Data availability

The ReliefNet web application and demonstration materials are publicly accessible at: Web Platform: <https://disasterseverity.netlify.app/> Demonstration Video: https://drive.google.com/file/d/1NB3iAfJRvainjyT_yD4z-9uiCyUi5e68/view?usp=drive_link

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